

IS5109 – IS Project for Community



Web-Based Public Complaint Management System for Pradeshiya Sabha

Group 17

Department of Computing & Information Systems

Faculty of Computing

Sabaragamuwa University of Sri Lanka

September - 2025

Approval of Community Project

Index Number	Name with Initials	Email	Mobile No
21CIS0001	M.I.Nuhas	minahamed@std.foc.sab.ac.lk	0764376883
21CIS0002	S.Imdaad	simdaad@std.foc.sab.ac.lk	0754738475
21CIS0003	R.M Ishan	rmishan@std.foc.sab.ac.lk	0727578276
21CIS0078	S.Theyshne	tsriharan@std.foc.sab.ac.lk	0763708118
21CIS141	J. Sujanthini	sjeyasingam@std.foc.sab.ac.lk	0772643951

1. Title of the Project:

Web-Based Public Complaint Management System for Pradeshiya Sabha

2. Name of the Mentor: Ahamed Rujhan

3. Mentor's designation: Associate Software Engineer at Omobio

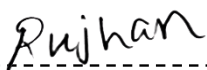
4. Name of the Internal Supervisor: Mrs. P.M.D.K. Kaushalya

5. Internal Supervisor's designation: Lecturer (Probationary),
Department of Computing & Information System,
Faculty of Computing, SUSL

For office use only:

Approved/Not approved

Signature of the Mentor



Date: 29.09.2025

Suggestions if any:

Table of Content

1. Acknowledgment.....	4
2. Introduction and Objectives	5
3. Analysis	6
3.1 Feasibility study	6
3.1.1 Technical Feasibility	6
3.1.2 Operational Feasibility	6
3.1.3 Economic Feasibility	6
3.2 Diagrams	7
3.2.1 Use Case Diagram	7
3.2.2 ER Diagram	8
3.2.3 DFD Diagram	9 - 10
4. Developing Environment	11
4.1 Hardware requirements.....	11
4.2. Software Requirements.....	11
4.3 Technology requirements	11
5. Tables and Structure.....	12
5.1 Number of Modules.....	12
5.2 Module Details	12
6. Proposed System	13
6.1 Requirements	13
6.1.1 Functional Requirements	13
6.1.2 Non-Functional Requirements	13
6.1.3 Security Requirements	13
6.2 Uniqueness of the Product	14
6.3 Methodology (Agile Methodology)	14
7. Gantt Chart	15
8. Cost Estimate	16
9. Conclusion	17
10. References	17
11. Appendix	18

1. Acknowledgment

We extend our sincere gratitude to Prof. S. Vasanthapriyan, Dean of the Faculty of Computing at Sabaragamuwa University of Sri Lanka, for his invaluable guidance and support in steering us in the right direction. We are also deeply thankful to Dr. L. S. Lekamge, Head of the Department of Computing, for his timely and insightful guidance.

Our heartfelt appreciation goes to Mrs. K.G. L. Chathumini, Lecturer (Probationary) and Coordinator of the Community Project, Department of Computing and Information Systems, Faculty of Computing, for her direction and support throughout this project. We would like to express our sincere gratitude to our Internal Supervisor, Ms. P.M.D.K. Kaushalya, for her invaluable guidance, support, and encouragement throughout this project.

We also extend our thanks to the Department of Computing & Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka, for providing us with the necessary resources and opportunity to undertake this community project. We would also like to express our sincere thanks to Addalaichenai Pradeshiya Sabha for their trust and collaboration. Their input and cooperation have been crucial in shaping this project, and we are grateful for the opportunity to contribute to enhancing their online presence.

2. Introduction and Objectives

2.1 Introduction

Efficient handling of public complaints is vital for maintaining community satisfaction and transparency in local government services. Many Pradeshiya Sabhas still rely on manual or paper-based processes, which delay response times and make tracking difficult.

This project proposes an **Online Complaint Management System** that enables citizens to submit, track, and receive feedback on complaints through a single platform. The system will categorize complaints into:

1. Waste Management Issues
2. Drainage-Related Issues
3. Road & Constructions
4. Streetlamp not working
5. Other Issues

By digitizing the process, the platform aims to increase accountability, provide faster resolutions, and improve communication between the Pradeshiya Sabha and the public.

2.2 Objectives

- To develop a secure and responsive web application for complaint registration and management.
- To provide a public interface for citizens to register, log in, and submit complaints with details such as title, description, address, location pin, and photo evidence.
- To create an administrative dashboard for Sabha officers to view, filter (by date, category, status), and manage complaints.
- To implement a communication channel (feedback/enquiry system) between admins and complainants.
- To ensure the system is user-friendly, accessible, and cost-effective for deployment.

3. Analysis

3.1 Feasibility Study

3.1.1 Technical Feasibility

- Frontend: Next.js 14+ for responsive UI for citizens and admins.
- Backend: next.js API routes for faster development, built-in serverless support, seamless integration, and performance optimization.
- Database: MongoDB Astro assess if the NoSQL database suits our need for flexible, dynamic data storage.
- Hosting: Netlify/Vercel (frontend), Render/Heroku for backend

3.1.2 Operational Feasibility

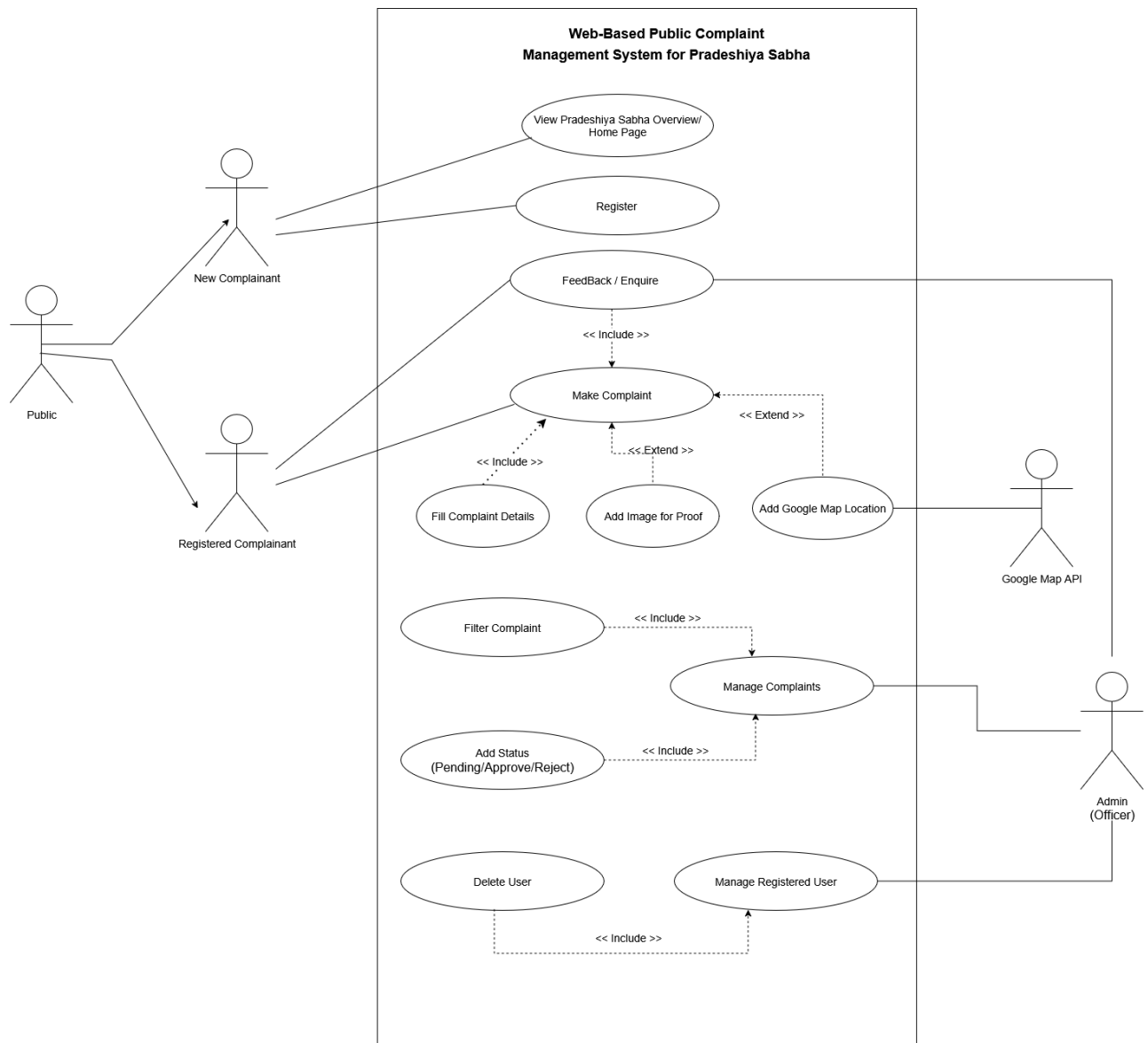
The system is highly operationally feasible. It simplifies the existing workflow for both the public and the Sabha officers. The interface is designed to be intuitive, requiring minimal training. By moving the process online, it reduces paperwork, saves time, and makes information easily accessible, leading to higher satisfaction and operational efficiency.

3.1.3 Economic Feasibility

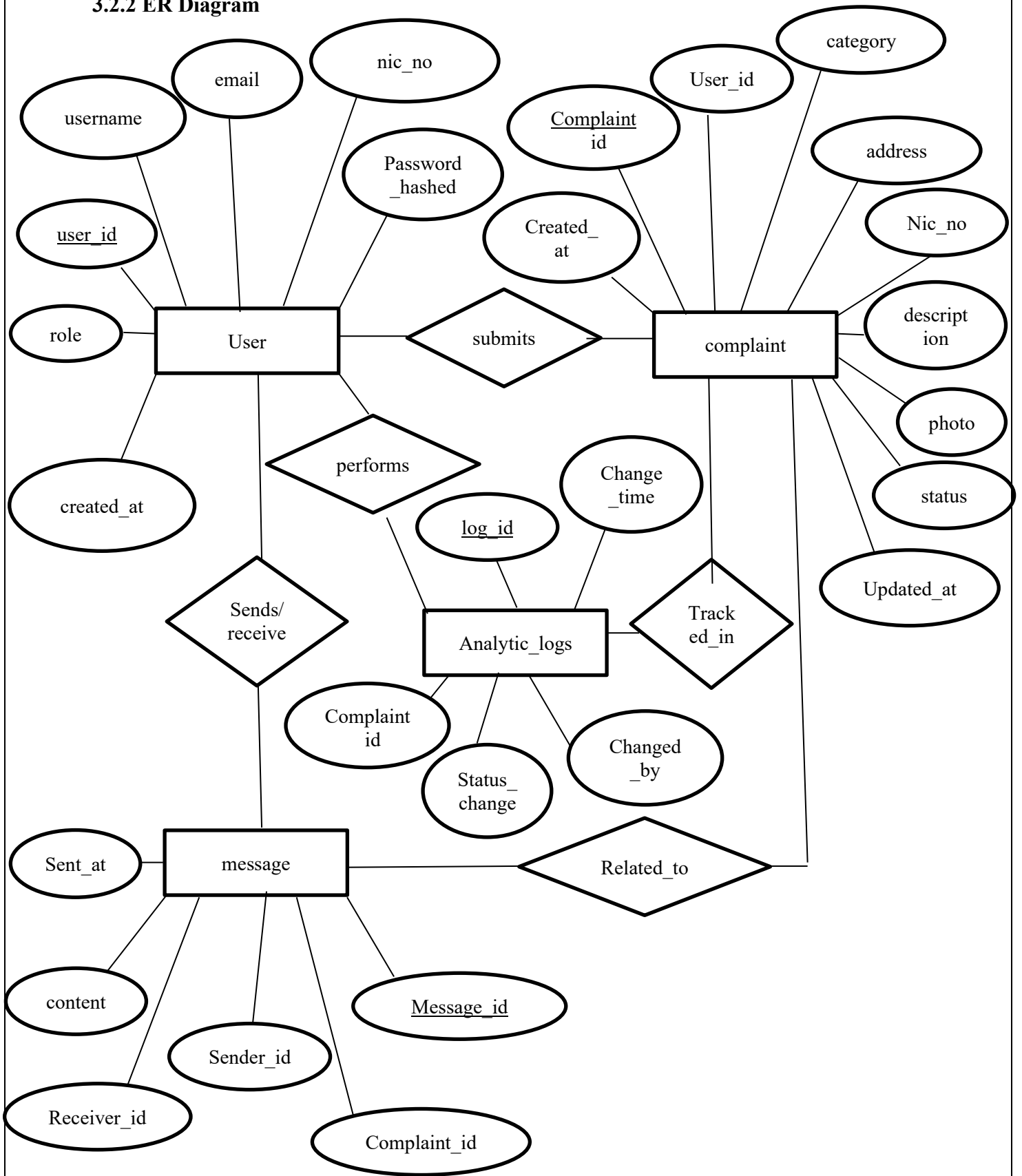
The project is economically feasible. The development uses free, open-source technologies. The estimated annual operational cost for hosting and database services is very low (approximately LKR 12,000 - 15,000 per year), which is a negligible expense for a government body compared to the significant improvements in service delivery and efficiency it offers.

3.2 Diagrams

3.2.1 Use Case Diagram

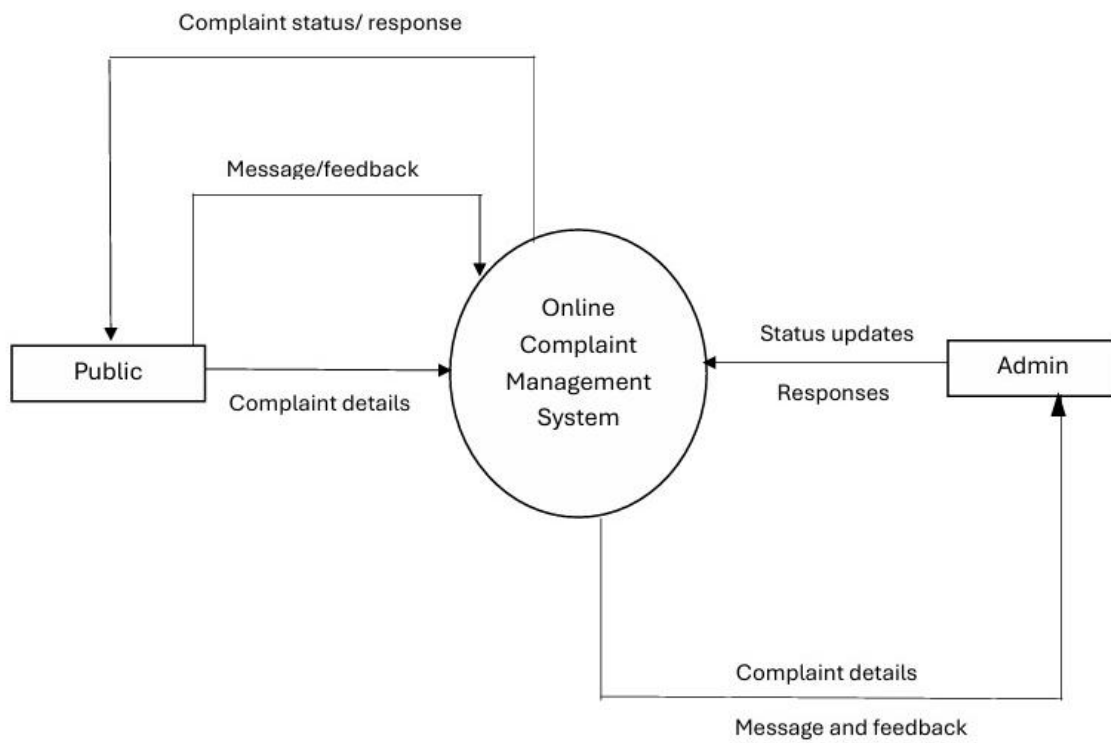


3.2.2 ER Diagram

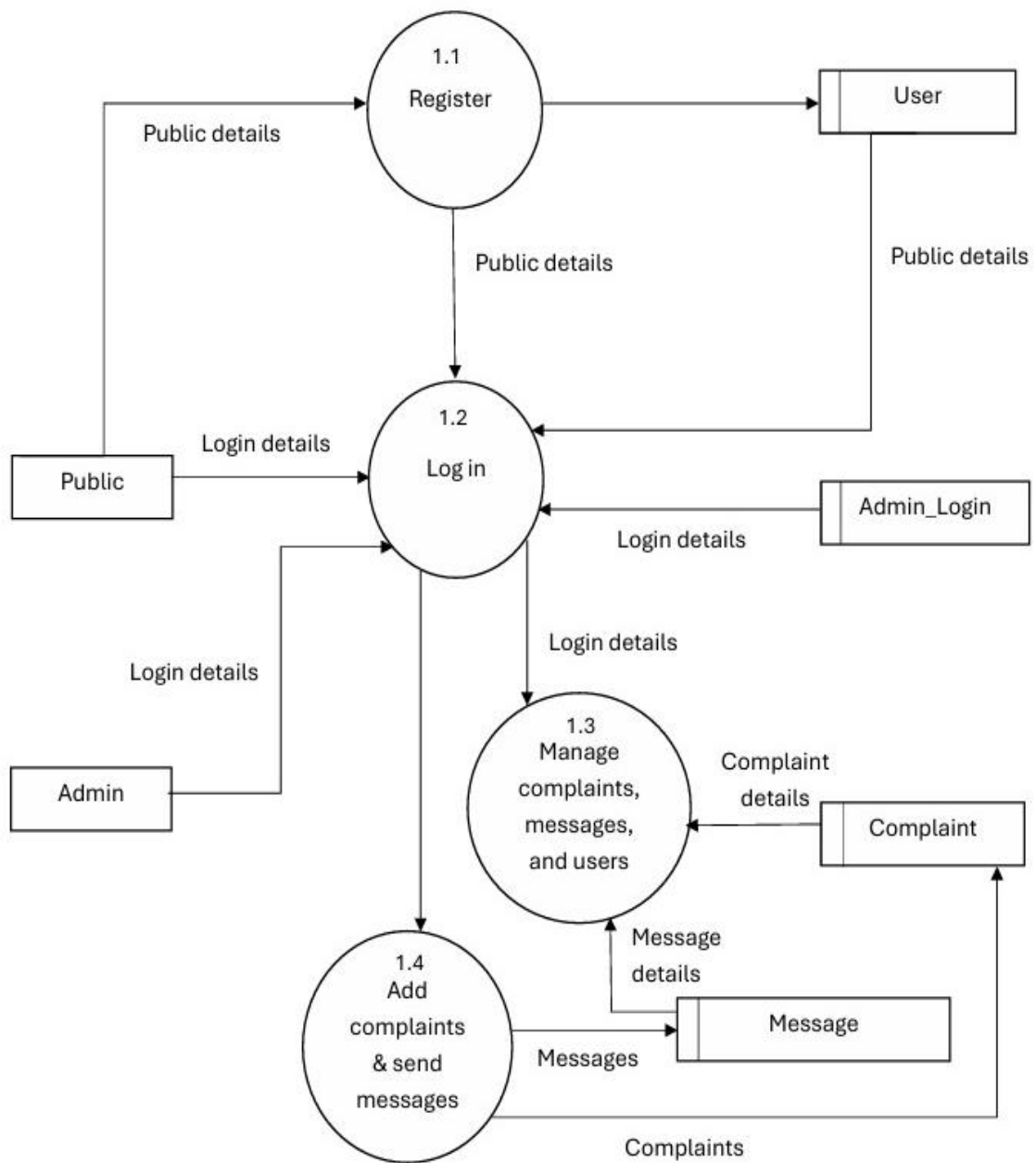


3.2.3 Data Flow Diagram (DFD)

DFD Level-0



DFD Level-1



4 Developing Environment

4.2 Hardware requirements

4.1.1 Development: Personal Computers/Laptops with minimum 4GB RAM, Core i3 processor or equivalent.

4.1.2 Production: Cloud server (VPS) specifications as provided by the hosting provider (e.g., 1 vCPU, 1GB RAM)

4.2. Software Requirements

- **OS:** Windows 10/11, macOS, or Linux.
- **Code Editor:** Visual Studio Code.
- **Version Control:** Git.
- **Design:** Figma.
- **Diagramming:** [Draw.io](https://draw.io).

4.3 Technology requirements

- **Frontend & Backend:** Next.js 14+ (React-based framework).
- **Styling:** Tailwind CSS.
- **Authentication:** JWT (JSON Web Tokens).
- **Database:** MongoDB (using Mongoose ODM).
- **Deployment Platform:** Vercel (optimized for Next.js) or a VPS like Hostinger/DigitalOcean.
- **Database Hosting:** MongoDB Atlas (cloud database service).

5 Tables and Structure

5.1 Number of Modules

The system will be structured into two main modules:

1. Public User Module
2. Admin Module

5.2 Module Details

- Public User Module:
 - Home Page (Overview of Sabha services)
 - User Registration & Login
 - Complaint Submission Form
 - Complaint Status Viewing
 - Feedback/Enquiry Interface
- Admin Module:
 - Admin Login
 - Dashboard with Complaint Overview
 - Complaint Management (View, Filter, Update Status, Resolve)
 - User Management (View Registered Users, Delete if necessary)
 - Feedback/Response System

6 Proposed System

6.1 Requirements

6.1.1 Functional Requirements

1. The system shall allow new users to register by providing necessary details.
2. The system shall allow registered users and admins to log in securely.
3. Registered users shall be able to submit a complaint by selecting one of five main categories: Drainage Issues, Street Lamp Not Working Issues, Waste Management Related Issues, Road and Construction Related Issues, and Other Issues.
4. Registered users shall be able to submit a complaint with details including title, description, address, location (optional), and photos (optional).
5. Admins shall be able to view all complaints in a dashboard.
6. Admins shall be able to filter complaints by category, date, and status.
7. Admins shall be able to update the status of a complaint (e.g., Pending, In-Progress, Resolved).
8. Both users and admins shall be able to exchange messages/feedback regarding a specific complaint.

6.1.2 Non-Functional Requirements

1. Usability: The interface shall be intuitive and easy to use for people with basic computer skills.
2. Performance: The web pages should load within 3 seconds under normal network conditions.
3. Availability: The system should aim for 99% uptime.
4. Security: User passwords shall be hashed, and data transmission shall be secured via HTTPS.

6.1.3 Security Requirements

1. User authentication shall be implemented using JWT.
2. All user inputs shall be validated and sanitized to prevent SQL injection and XSS attacks.
3. API routes shall be protected; access to admin routes shall be restricted to users with admin roles.
4. Sensitive data like passwords shall be hashed using bcrypt before storage.

6.2 Uniqueness of the Product

The system promotes transparency and accountability by allowing citizens to submit complaints digitally and track their real-time status, minimizing the risk of unresolved or ignored issues. Moreover, the integrated feedback and messaging feature enables two-way communication between the public and administrators, fostering trust and responsiveness in local governance. By combining a user-friendly public portal with a powerful admin dashboard, the system unifies the complaint-handling process within a single platform, streamlining operations for both citizens and officials. Additionally, it introduces a shift from traditional paper-based methods to a modern digital platform, saving time, improving efficiency, and ensuring that no complaint goes unnoticed. Its scalable design also makes it future-ready, with the potential to integrate SMS/email notifications, mobile application support, and analytical tools for data-driven decision-making. This community-focused, transparent, and interactive approach makes our solution unique and valuable to both the Pradeshiya Sabha and the citizens it serves.

6.3 Methodology (Agile Methodology)

We will adopt the Agile methodology for this project. Development will be iterative, divided into small sprints (each lasting 1-2 weeks). This approach allows for regular feedback, adaptability to changes, and continuous improvement. Key activities in each sprint will include planning, development, testing, and review.

7 Gantt Chart

Task	Time Duration (In weeks)														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Project Selection	■	■	■												
Project Planning				■											
Requirement Analysis & Definition					■	■									
Learning Techniques	■	■	■	■	■	■	■	■	■	■	■	■			
Documentation								■							
Design					■	■	■	■	■						
Development						■	■	■	■	■	■	■	■		
Test & Correction											■	■	■	■	
Implementation													■	■	
Project Completion														■	■

8 Cost Estimate

Item	Description	Estimated Cost (LKR) / Year	Notes
Domain Name	.com.lk or .lk domain (e.g., complaints.shabaname.lk)	~ LKR 2,500	From a registrar like LK Domain Registry or Hostinger.
Web Hosting	VPS (Virtual Private Server) Plan	~ LKR 7,200	E.g., Hostinger's VPS Startup plan (LKR 600/month). Vercel might have a free tier suitable for initial deployment.
Database Hosting	MongoDB Atlas (M0 Free Tier)	LKR 0	Free tier provides 512MB storage, sufficient for a project of this scale. If more is needed, paid tiers start at ~LKR 5,000/year.
SSL Certificate	For HTTPS security	LKR 0	Usually provided for free by hosting providers like Vercel/Hostinger.
Miscellaneous		LKR 1,000	Contingency for other small costs.
Total Estimated Annual Cost		LKR 10,700 - LKR 15,700	This is a very affordable and realistic cost .

9 Conclusion

The proposed Web-Based Public Complaint Management System is a viable, necessary, and cost-effective solution to modernize the operations of a Pradeshiya Sabha. By leveraging modern web technologies, the project aims to enhance civic engagement, improve the efficiency of public service delivery, and bring transparency to the complaint resolution process. The successful implementation of this system will benefit both the citizens and the Sabha officials, making it a valuable community project .

10 References

1. Next.js Documentation. <https://nextjs.org/docs>
2. MongoDB Atlas Documentation. <https://www.mongodb.com/docs/atlas/>
3. Tailwind CSS Documentation. <https://tailwindcss.com/docs>
4. JSON Web Tokens (JWT) Introduction. <https://jwt.io/introduction>
5. Vercel Deployment Documentation. <https://vercel.com/docs>
6. GitHub Documentation. <https://docs.github.com/en>

11 Appendix



அட்டாளைச்சேனை பிரதேச சபை
අඩලාචනවෙත ප්‍රදේශීය සභාව
Addalaichenai Pradeshiya Sabha

My No : AD/PS/RTI/GEN/2025 Date: 2025.09.18

Community Project Group 17,
Faculty of Computing,
Sabaragamuwa University of Sri Lanka

Approval for Community Project – Complaint Management Website

Dear Students,

We acknowledge the proposal submitted by your team regarding the development of a web-based Complaint Management System as part of your community project.

We are pleased to grant approval for you to carry out this project for the benefit of the people in our area. This system will support the Addalaichenai Pradeshiya Sabha in handling public complaints more efficiently and transparently.

In particular, please ensure the system includes the following:

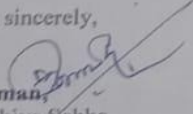
1. Facilities for the public to submit complaints related to roads, waste disposal, street lights, and other relevant services.
2. A facility for citizens to track the status of their complaints (Pending / Approved / Rejected).
3. A feedback or enquiry feature to improve communication between officers and the public.
4. An option for users to provide photographic proof or a Google Maps location of the complaint area, to enable easier tracking and greater accuracy (optional but recommended).

We look forward to your successful implementation of this project, which will bring great value to the community and strengthen service delivery.

This letter serves as the formal approval and request from the Addalaichenai Pradeshiya Sabha for your group to proceed with the project.

Thank you for your initiative and contribution.

Yours sincerely,



Chairman,
Pradeshiya Sabha,
Addalaichenai

AS. MOHAMEDU UWAISS
Chairman
Pradeshiya Sabha
Addalaichenai

Jalaldeen Road, Addalaichenai

 addalaichennai.ps.gov.lk

 +94765046231

 Addalaichenai Pradeshiya Sabha

 067-2277375

 067-2278600

 adpslg24@gmail.com